

Real-Time UWB-IMU Factor-Graph Fusion with Integrity Monitoring

Architecture, Validation, and Full-History vs. Fixed-Lag Analysis

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Artifact Git 0ed90a0d32c2; report protocol d3d0fa4f3cd7

Problem statement: accuracy is not enough

Navigation objective

Estimate pose, velocity, and inertial bias at 20 Hz from 200 Hz IMU and grouped UWB ranges.

Integrity objective

Detect a bad current range group before it contaminates the graph, and report protection levels that bound position error under the declared fault model.

$$\text{measurement} \rightarrow \begin{cases} \text{commit}, & T \leq \gamma \\ \text{reject}, & T > \gamma \end{cases}$$

Accuracy $\neq$ consistency $\neq$ integrity

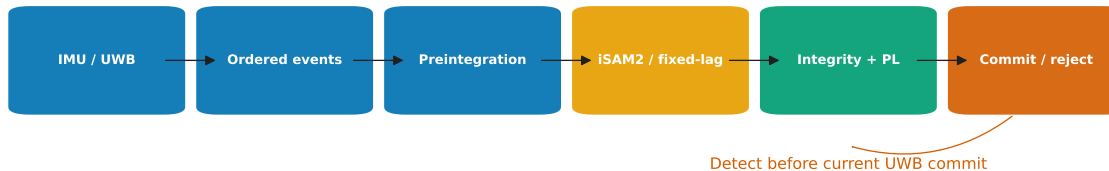
Current development status

Implemented	Formally/component validated	Not yet closed
Incremental iSAM2; fixed-lag marginalization; combined IMU and UWB factors; detect-before-commit; current-group PL	Noncentral calculation; IMU MC; Method A/B shadow; 8 ROS topic scenarios; fixed-lag performance	Week-4 Snapshot/ROC/History gates; real data; certified persistent-fault PL; FDE; multiple faults; rare-event certification

Authoritative status

Frozen Week-4 acceptance remains **INVALID**. Advisor-report experiments are a separate evidence namespace and cannot change it.

End-to-end architecture

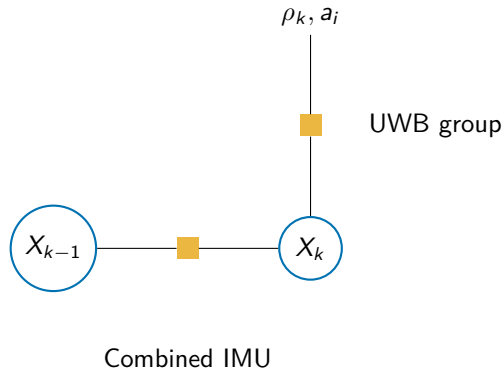


Callback work is enqueue-only; one ordered worker owns state mutation and logging.

State and factor graph

$$\mathcal{X}_k = \{R_k, p_k, v_k, b_k^a, b_k^g\}$$

- Pose prior, velocity prior, and bias prior at $k = 0$.
- Combined IMU factor links $(k - 1, k)$.
- One grouped UWB factor preserves the full range covariance.
- iSAM2 maintains a Bayes tree; only the current clique closure is queried.



IMU preintegration

$$\begin{aligned}\Delta \tilde{R}_{ij} &= \prod_{k=i}^{j-1} \text{Exp}((\tilde{\omega}_k - b_i^g) \Delta t), \\ \Delta \tilde{v}_{ij} &= \sum_{k=i}^{j-1} \Delta \tilde{R}_{ik} (\tilde{a}_k - b_i^a) \Delta t, \\ \Delta \tilde{p}_{ij} &= \sum_{k=i}^{j-1} \left[\Delta \tilde{v}_{ik} \Delta t + \frac{1}{2} \Delta \tilde{R}_{ik} (\tilde{a}_k - b_i^a) \Delta t^2 \right].\end{aligned}$$

Bias correction is first order:

$$\Delta v(b_i + \delta b) \simeq \Delta \tilde{v} + J_{v,b} \delta b.$$

Error covariance propagates with the configured accelerometer, gyroscope, and bias random-walk densities.

UWB measurement model and provenance

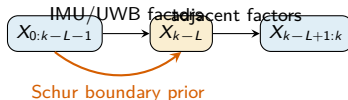
$$r_{k,i} = \|p_k + R_k \ell - a_i\| - \rho_{k,i}$$

- Physical anchor ID, world-frame coordinate, map version, measurement ID, and factor ID are retained through the integrity snapshot.
- For grouped covariance $\Sigma_\rho = LL^\top$, rows are whitened by L^{-1} :

$$e_w = L^{-1}e, \quad H_w = L^{-1}H.$$

- Cholesky whitening is audited numerically; malformed or non-SPD covariance fails closed.
- No trajectory alignment is applied when computing errors: truth and estimate already share the world frame.

Two incremental backend modes



Property	full-history incremental iSAM2	fixed-lag incremental smoother ($L = 200$)
History representation	Explicit states and factors from epoch 0	Retained window plus a Schur boundary prior
Relinearization scope	Affected Bayes-tree cliques; potential scope grows	Affected retained cliques and dense boundary factor
Graph / memory	Grows with sequence length	Bounded after the first marginalization
Extra work	No fixed-lag marginalization	Eliminate expired keys; update boundary prior
Best use	Short runs, reference solution, offline smoothing	Long-running bounded-resource navigation

Both are incremental; “fixed lag” changes history representation, not the current measurement model.

Detect before commit

predict $\rightarrow$ pre-UWB prior $\rightarrow$ integrity test $\rightarrow$ commit / reject

- 1 IMU/history creates the current state key.
- 2 Extract P^- and linearize the proposed UWB group at one consistent point.
- 3 Evaluate detector and protection level from a read-only snapshot.
- 4 Commit only if the current group passes the model and detector gates.

Invariant

The current UWB group must not appear in P^- ; graph version, ordering version, row provenance, and pending batch ID are audited.

Conditional current-group innovation detector

$$\begin{aligned}S_w &= H_w P^- H_w^\top + I, \\T &= e_w^\top S_w^{-1} e_w \sim \chi_m^2 \quad (H_0), \\ \gamma &= F_{\chi_m^2}^{-1}(1 - P_{FA}).\end{aligned}$$

- m is the number of ranges in the current group; here $m = 8$ nominally.
- Global graph residual and current post-fit residual are diagnostics requiring independent empirical calibration.
- The conditional statistic is the primary detector because its prior explicitly excludes the tested group.

Protection level and declared scope

$$\begin{aligned}P^+ &= P^- - P^- H_w^\top S_w^{-1} H_w P^-, \\PL_j &= K_{\text{nom}} \sigma_j + \max_i g_{i,j} \sqrt{\bar{\eta}_i}, \\HPL &= \sqrt{PL_x^2 + PL_y^2}, \quad VPL = PL_z.\end{aligned}$$

- One hypothesis per physical anchor in the current UWB group.
- $\bar{\eta}_i$ comes from the noncentral χ^2 missed-detection allocation.
- Availability also requires valid geometry, valid total risk allocation, and HPL/VPL below alert limits.

Formal boundary

Current-group, single-anchor range bias only. Historical persistent faults are not represented by the present fixed-lag boundary prior provenance.

Research questions

- 1 Does a 10 s fixed lag preserve full-history accuracy, covariance, and PL behavior?
- 2 Does fixed lag bound graph size, memory, and tail latency in practice?
- 3 Are conditional detector PFA, power, and time-to-detect comparable between modes?
- 4 Which claims are supported by deterministic simulation, and which still require formal or real-data validation?

Simulation setup

- Trajectories: straight, circle, figure-eight.
- Eight 3-D anchors at cube corners $x, y = \pm 5$ m, $z \in \{1, 5\}$ m.
- IMU 200 Hz; grouped UWB 20 Hz.
- Noise and bias random walks come from the sole research YAML.
- Same generated input object is replayed to every backend mode; digest is checked per sequence.

Interpretation

High-fidelity deterministic simulation and component evidence, not measured-data validation. Errors are computed directly in the common world frame; no Umeyama alignment.

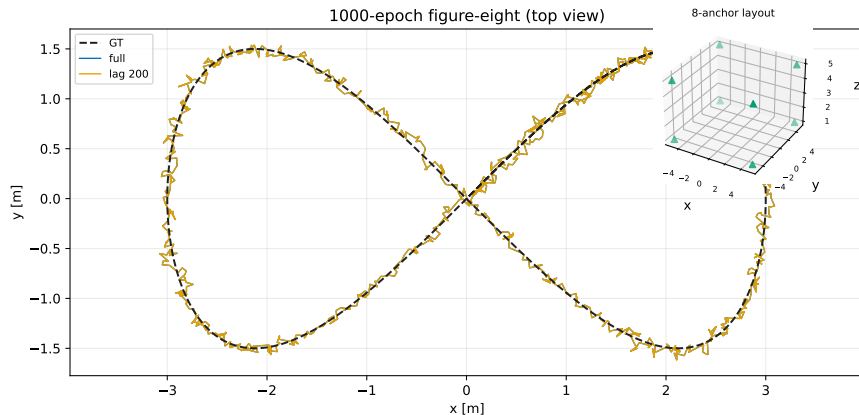
Focused-PL experiment matrix

Suite	Modes	Seeds	Traj.	Epochs/seq.	Profile / purpose
Nominal paired	2	2	3	250	smoke accuracy / NEES
Window ablation	5	1	3	250	smoke trade-off
ROC calibration/eval.	2	reduced	3	300	smoke detector comparison
Performance	2	1	1	500	smoke latency / RSS
Isolated fault	2	640	3	312	complete PL challenge
Persistent 1 m fault	2	80	3	600	complete policy invariant
Representative path	2	1	1	1000	complete figure-eight

- Isolated matrix: 8 anchors $\times$ 8 fixed magnitudes $\times$ 10 seeds; seed families are disjoint.
- Fault onset is epoch 300; only requested neighborhoods are saved, while every preceding epoch executes.
- The two modes replay the identical generated IMU/UWB object and must share one input digest.

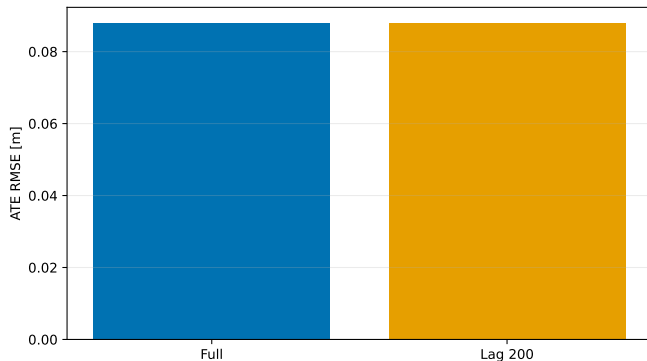
FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

Representative trajectories and anchor layout



FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

Nominal position accuracy



FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

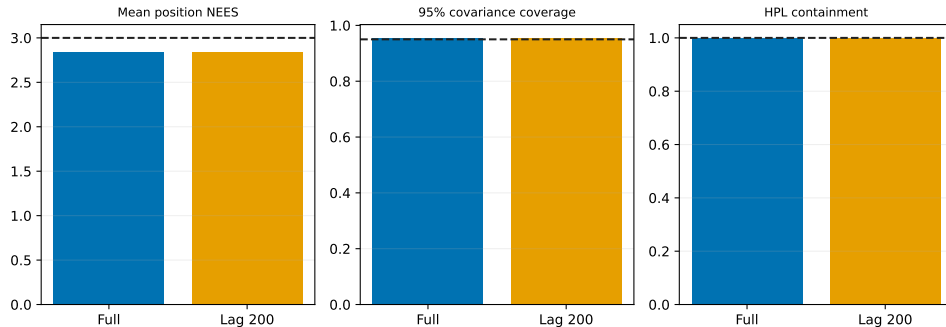
	Full	Lag 200
ATE RMSE [m]	0.088	0.088
P99 [m]	0.184	0.184
Mean core [ms]	1.47	2.40

Paired blocks: 6. CI upper degradation:

0.00%. Criterion: PASS.

Current graphs contain only adjacent IMU factors and the current grouped UWB factor. Marginalizing old states transfers their information to the retained boundary through the Schur prior; therefore the current state can be numerically consistent across modes in this setup.

Consistency and integrity containment



NEES: full 2.842, lag 2.842.

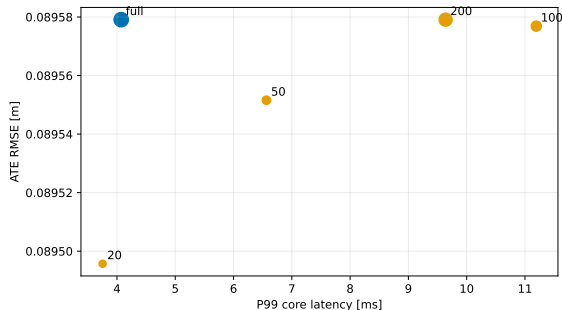
Mean position

Negative result retained

Orientation RMSE is full 1.710 rad and lag 1.710 rad; yaw is weakly observable in this identity-attitude, range-only simulation. Position ATE must not be used to claim full navigation-state accuracy.

FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

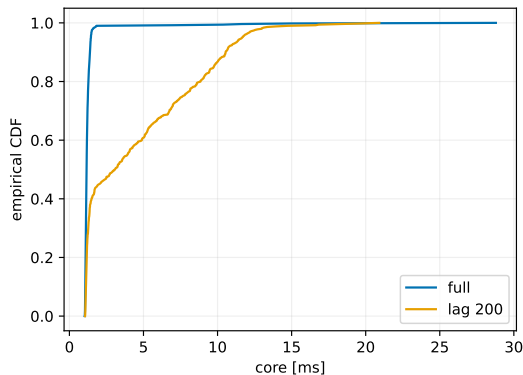
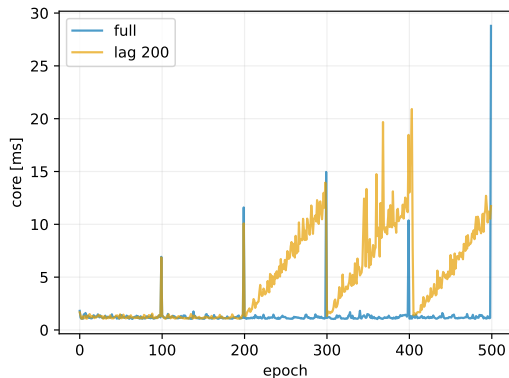
Window-length ablation



Marker size encodes maximum active factor count; labels are retained history lengths in epochs. A scientific trade-off is reported only after the full seed matrix is complete.

FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

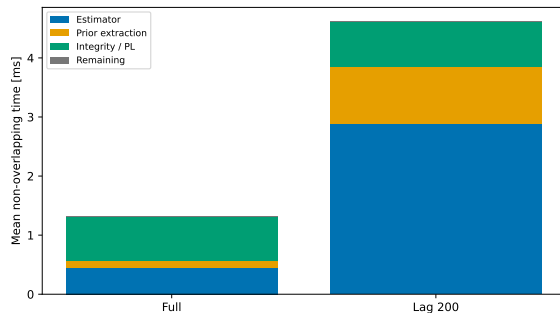
Latency comparison



This is a 500-epoch smoke comparison. At this length the fixed-lag path pays boundary marginalization and prior-extraction cost, which can dominate any saving from a bounded graph. “Bounded size” does not mean “faster at every sequence length.” Reference lines are 50 ms mean and 100 ms P99; these data are not a platform gate.

FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

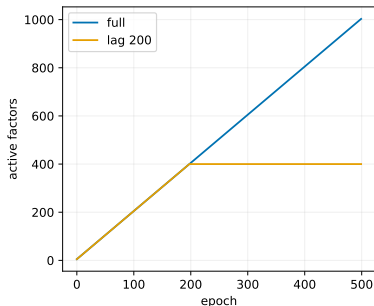
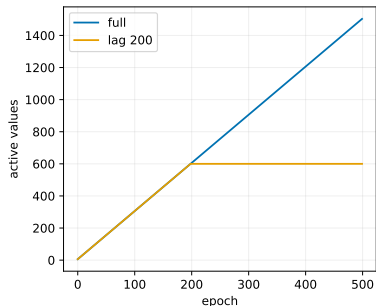
Non-overlapping timing breakdown



Categories partition core wall time: estimator update/commit, prior marginal and snapshot extraction, integrity/PL stages, and remaining overhead. Nested sub-timers are not summed twice.

FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

Resource growth and marginalization



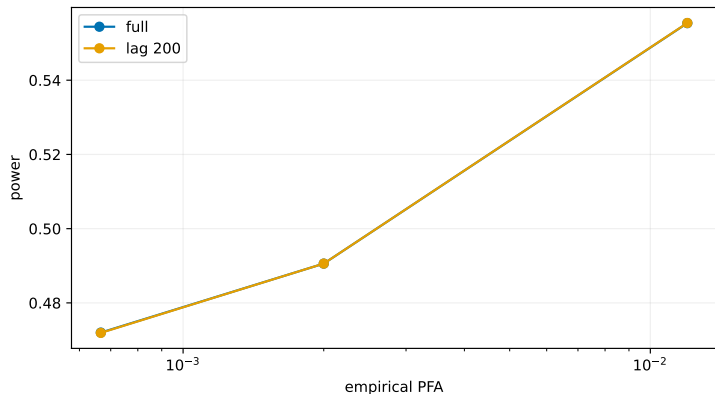
Maximum active factors in current nominal evidence:

- full-history: 503
- fixed-lag: 401

Full-history growth is expected to continue; the 200-epoch smoother becomes bounded after marginalization begins.

FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

Detector ROC and AUC



Conditional AUC:

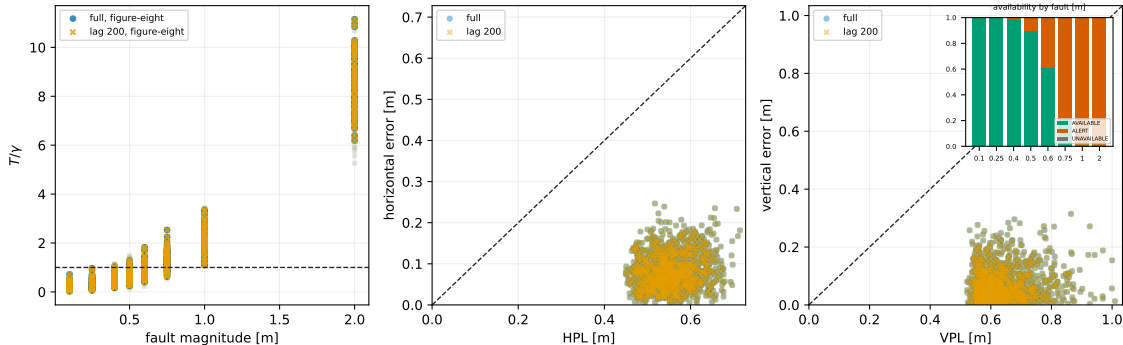
- Full: 0.846
- Lag 200: 0.846

Calibration, nominal evaluation, and fault seeds are disjoint. 10^{-5} is theory-only; finite samples cannot certify rare events.

Smoke Holm PFA check: **FAIL**;
conditional power NI check: **PASS**.
Both are reported, but neither upgrades the formal gate.

FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

Single-epoch fault: empirical PL challenge

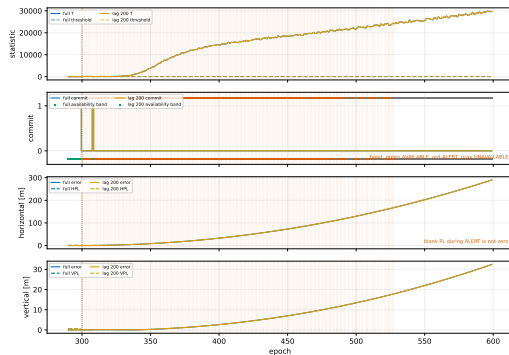


Left: $T/\gamma > 1$ is rejected. Middle/right: points below the $error = PL$ diagonal are contained; the inset reports AVAILABLE/ALERT/UNAVAILABLE fractions by magnitude. Figure-eight points are emphasized; other trajectories remain visible in pale color.

Finite-PL denominator: 2252 / 3840; containment: 100.00%; near-boundary samples ($0.8 \leq T/\gamma \leq 1.2$): 584. Observed HMI: 0; one-sided exact 95% upper rate: 0.078%. Negative points, if any, are retained and counted.

FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

Persistent 1 m fault: conservative service withdrawal



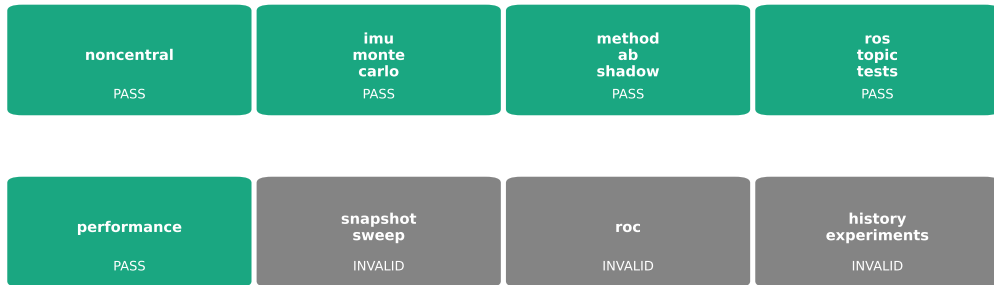
Across 8 anchors, 10 seeds/anchor, and 3 trajectories:
Rejection 98.94%; longest run 300 epochs;
ALERT / UNAVAILABLE 70.72% /
28.22%; max IMU-only drift 2036.528 m;
rejected rows with finite PL: 0.
Blank HPL/VPL in a red ALERT band
means the integrity service was
withdrawn—not zero PL and not missing
plotting data.

Claim boundary

The evidence supports detection followed by integrity-service withdrawal. It is not a formal certification of PL under persistent faults.

FOCUSED_PL: complete focused PL matrices; baseline accuracy/performance/ROC remain smoke comparisons.

Validation dashboard: frozen evidence, not upgraded



Existing checksums were re-verified before reuse. Formal PASS: Noncentral, IMU MC, Method A/B shadow, ROS topic tests, and fixed-lag performance. Snapshot/ROC/History remain **INVALID** in frozen Week-4 acceptance.

Conclusions and next steps

Supported now

Both modes are incremental iSAM2 backends; fixed-lag uses a Schur boundary prior. The implementation enforces detect-before-commit and paired input identity. The focused challenge reports every observed HMI and confirms that rejected batches publish no finite PL.

Not supported yet

Baseline comparisons remain smoke; orientation observability is weak; Week-4 Snapshot/ROC/History are not closed. No claim is made for real data, certified persistent-fault PL, FDE, multiple faults, TDoA, rare events, or nonlinear remainder certification.

Next: add measured datasets and an outage/IMU-drift integrity model with persistent-fault provenance or FDE; then run rare-event importance sampling without changing the fixed seeds.

Backup: conditional innovation derivation

Given $\delta x \sim \mathcal{N}(0, P^-)$ and $e_w = H_w \delta x + \nu_w$, $\nu_w \sim \mathcal{N}(0, I)$,

$$e_w \sim \mathcal{N}(0, S_w), \quad S_w = H_w P^- H_w^\top + I.$$

Let $z = S_w^{-1/2} e_w$. Under H_0 , $z \sim \mathcal{N}(0, I_m)$, hence

$$T = e_w^\top S_w^{-1} e_w = z^\top z \sim \chi_m^2.$$

For anchor fault incidence a_i , the noncentrality is $\eta_i = f_i^2 a_i^\top S_w^{-1} a_i$.

Backup: PL ingredients and risk allocation

$$K = P^- H_w^\top S_w^{-1},$$
$$g_{i,j} = \frac{|e_j^\top J_p K a_i|}{\sqrt{a_i^\top S_w^{-1} a_i}},$$

$$P^+ = (I - K H_w) P^-,$$

$$P(HMI) \leq P_{NM} + \sum_i P(H_i) P_{MD,i}.$$

The implementation checks: complete physical anchor map, monitored hypotheses, finite sensitivity, total allocated risk, current linearization version, and alert limits. Any failed prerequisite returns unavailable rather than a finite formal PL.

Backup: resolved research parameters

Parameter	Value	Parameter	Value
IMU rate	200 Hz	UWB rate	20 Hz
σ_a	0.10	σ_ω	0.01
σ_{b_a}	0.001	σ_{b_g}	0.0001
Range σ	0.10 m	Gravity	9.80665 m/s ²
P_{FA} (formal config)	10 ⁻⁵	Total P_{HMI}	4 × 10 ⁻⁵
Single-anchor P_{MD}	10 ⁻³	Fixed lag	200 epochs
HAL / VAL	2 / 3 m	iSAM2 relin. skip	200 updates

Values are loaded from the resolved research configuration; the

report protocol defines only matrices, seeds, bootstrap, and display criteria.

Backup: statistical protocol

- Exact timestamp matching; no spatial or temporal alignment after estimation.
- Bootstrap unit is the full (*seed*, *trajectory*) sequence, not individual epochs.
- Calibration, nominal evaluation, and fault seed ranges are disjoint and checked.
- PFA comparisons use Holm $\text{FWER} = 0.05$ in the full campaign.
- TTD is the first threshold crossing after the onset inside one block; undetected sequences remain censored.
- Scientific failure is reported as FAIL. Missing, malformed, duplicate, non-finite, or incomplete artifacts are INVALID.

Backup: reused component evidence

Component	Frozen result	Scope	Source Git and protocol hash
Noncentral MC	PASS	Global bootstrap, Holm, H_0 checks	
IMU MC	PASS	3 scenarios $\times$ 100,000 trials	
Method A/B shadow	PASS	600/600 candidates	
ROS topic tests	PASS	8/8 deterministic scenarios	
Performance	PASS	Fixed lag, 3 $\times$ 12,000 epochs	

remain those of the frozen Week-4 artifact; checksum verification succeeds. These component results do not substitute for unfinished Snapshot/ROC/History gates.

Backup: Week-4 acceptance boundary

PASS

Noncentral, IMU Monte Carlo, Method A/B shadow, ROS topic tests, fixed-lag performance.

INVALID / incomplete

Snapshot sweep, ROC, history experiments. Therefore overall status is INVALID and no success token exists.

The advisor-report artifact records `week4_acceptance_modified=false`.

Backup: selected references



C. Forster et al., “On-Manifold Preintegration for Real-Time Visual–Inertial Odometry,” *IEEE TRO*, 2017.



M. Kaess et al., “iSAM2: Incremental Smoothing and Mapping Using the Bayes Tree,” *IJRR*, 2012.



GTSAM, Incremental Fixed-Lag Smoother documentation and implementation, version recorded in the run manifest.



R. Brown and G. Hwang, *Introduction to Random Signals and Applied Kalman Filtering*, integrity and consistency foundations.

All numerical claims in the main deck are generated from `summary.json`; no result value is hand-copied into this source.